D.C. Circuit Theory

Engineering Technology

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

D.C. Circuit Theory (A11848)

Short Title:	D.C. Circuit Theory		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Electrical Engineering		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This is the first time the student encounters Circuit Theory. The course introduces circuit analysis for resistive networks. It introduces capacitors & inductors and transient step function responses for RC & RL circuits. Ferromagnetism is explained and B-H curves treated. Teaching is carried out in the classroom and laboratory.

Programmes

	BEng (Hons) in Automation Engineering with Data Intelligence (`WD_EAUTO_B`)	1 / 2 / E
`ELTR-0012`	BEng (Hons) in Electrical Engineering (`WD_ETRIC_B`)	1 / 2 / E
	BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	1 / 2 / M
`ELTR-0012`	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	1 / 2 / E
	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	1 / 2 / M
`ELTR-0012`	BEng (Hons) in Sustainable Civil Engineering (`WD_ESCIV_B`)	1 / 2 / E
`ELTR-0012`	BEng (Hons) in Sustainable Energy Engineering (`WD_ESERG_B`)	1 / 2 / E
`ELTR-0012`	BEng (Hons) (`WD_ETRIC_B`)	1 / 2 / E

Indicative Content

- Resistance & simple circuits: resistivity, Ohm's Law, series & parallel, Kirchhoff's Laws, current & voltage divider laws
- Network Theorems: Thevenin & Norton equivalents, loop analysis, delta-star/star-delta conversion, maximum power transfer for resistive loads
- Electrostatics and Capacitance: permittivity, first order (RC) transients - solution of first order differential eqn, series & parallel
- Introduction to Magnetism and Inductance; permeability, B-H curves, ferromagnetism, domains, hysteresis, first order (RL) transients, series & parallel, Ampère's circuital law

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Analyse and explain the behaviour of simple circuits by employing fundamental circuit theory rules.
2. Examine and transform sources and circuit configurations, to enhance understanding and evaluation of a circuit.
3. Derive and apply the Maximum Power Transfer Theorem to any suitable circuit element.
4. Predict the step response of RC & RL circuits.
5. Interpret B-H curves for magnetic materials.
6. Demonstrate a practical ability to develop, troubleshoot, test, analyse and report on D.C circuits.

Learning and Teaching Methods

- Lecture / Lectorial
- Laboratory

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1, 2, 3 , 4 , 5
Continuous Assessment	30%	
In-Class Assessment	10%	1,2,3,4,5
Practical	20%	6

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Hughes, E., J. Hiley, K. Brown and I. McKenzie Smith. _Electrical and Electronic Technology_. 12th ed.. Harlow: Pearson, 2016.

Supplementary Material

- Irwin, D.J. and R.M. Nelms. _Engineering Circuit Analysis_. 10th Ed.. New York: John Wiley, ..